DBT PORTFOLIO PROJECT

LAYER: INTERMEDIATE

DBT Intermediate Model

Value at Risk | VaR | Risk Analytics | Statistics | Snowflake | Finance

Project Summary

Computes parametric Value at Risk (VaR) at three confidence levels (90%, 95%, 99%) for healthcare financial returns, segmented by entity, payer, state, and specialty. Uses the closed-form normal distribution formula (mu - z*sigma) with STDDEV_SAMP for unbiased standard deviation. Results feed directly into risk dashboards and financial planning reports.

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1 PROJECT OVERVIEW

Model Name:	int_value_at_risk
Layer:	Intermediate
Materialization:	table
VaR Formula:	mu - z * sigma (parametric normal distribution)
Confidence Levels:	90% (z=1.2816), 95% (z=1.6449), 99% (z=2.3263)
Grain:	One row per entity_id / payer / state / specialty
Audience:	Actuaries, CFO, Risk Management, Finance leadership

2 BUSINESS PROBLEM

Finance leadership needed to answer: 'What is the worst-case financial loss we should expect from our claims portfolio at a 95% confidence level?' VaR quantifies this in a single number -- the threshold loss that will only be exceeded 5% of the time. This model delivers VaR at three confidence levels for every entity/payer/state/specialty combination, enabling granular risk monitoring without any Python or actuarial software.

- var_90_pct: Maximum expected loss exceeded only 10% of the time.
- var_95_pct: Industry standard risk threshold -- exceeded only 5% of the time.
- var_99_pct: Stress scenario threshold -- exceeded only 1% of the time (tail risk).
- STDDEV_SAMP: Uses n-1 denominator (unbiased) rather than STDDEV_POP (n denominator).
- Grouped by payer, state, specialty: Enables drill-down risk analysis by segment.

3 TECHNICAL EXPLANATION

- stats CTE: Aggregates return_value by entity/payer/state/specialty -- computes AVG (mu), STDDEV_SAMP (sigma), and COUNT (n).
- Parametric VaR formula: VaR = mu - z * sigma where z is the standard normal critical value at each confidence level.
- z=1.2816 for 90% (one-tailed), z=1.6449 for 95%, z=2.3263 for 99%.
- Assumes returns are normally distributed -- appropriate for sufficiently large n (CLT). Flag in docs if n < 30.
- Materialized as TABLE: Mart layer and Sharpe ratio model both join to this -- table avoids redundant computation.

Key Techniques: AVG(), STDDEV_SAMP(), GROUP BY segmentation, parametric normal VaR formula, three confidence levels

4 FULL MODEL CODE

```
{{ config(materialized='table') }}

with stats as (
    select
        entity_id,
        payer,
        state,
        specialty,
        avg(return_value)        as mu,
        stddev_samp(return_value) as sigma,
        count(*)                  as n
    from {{ source('healthcare', 'claims') }}
    group by entity_id, payer, state, specialty
)

select
    entity_id,
    payer,
    state,
    specialty,
    mu,
    sigma,
    n,
    mu - 1.2816 * sigma as var_90_pct,
    mu - 1.6449 * sigma as var_95_pct,
    mu - 2.3263 * sigma as var_99_pct
from stats
```

5 EXAMPLE OUTPUT

entity_id	payer	state	mu	sigma	n	VaR 90%	VaR 95%	VaR 99%
E-001	BlueCross	AR	0.032	0.018	120	-0.0002	-0.0051	-0.0178
E-002	Aetna	TX	0.019	0.024	118	-0.0118	-0.0204	-0.0368
E-003	UnitedHlth	MO	0.041	0.012	122	0.0256	0.0213	0.0131

** Negative VaR indicates expected loss at that confidence level. Normal distribution assumed.*